

Who Is Left of Whom?

Tracing Spatial Evidence and Role Binding in Relative-Position Reasoning

Yingjin Song Denis Paperno Albert Gatt
Utrecht University, The Netherlands · [y.song5, d.paperno, a.gatt]@uu.nl

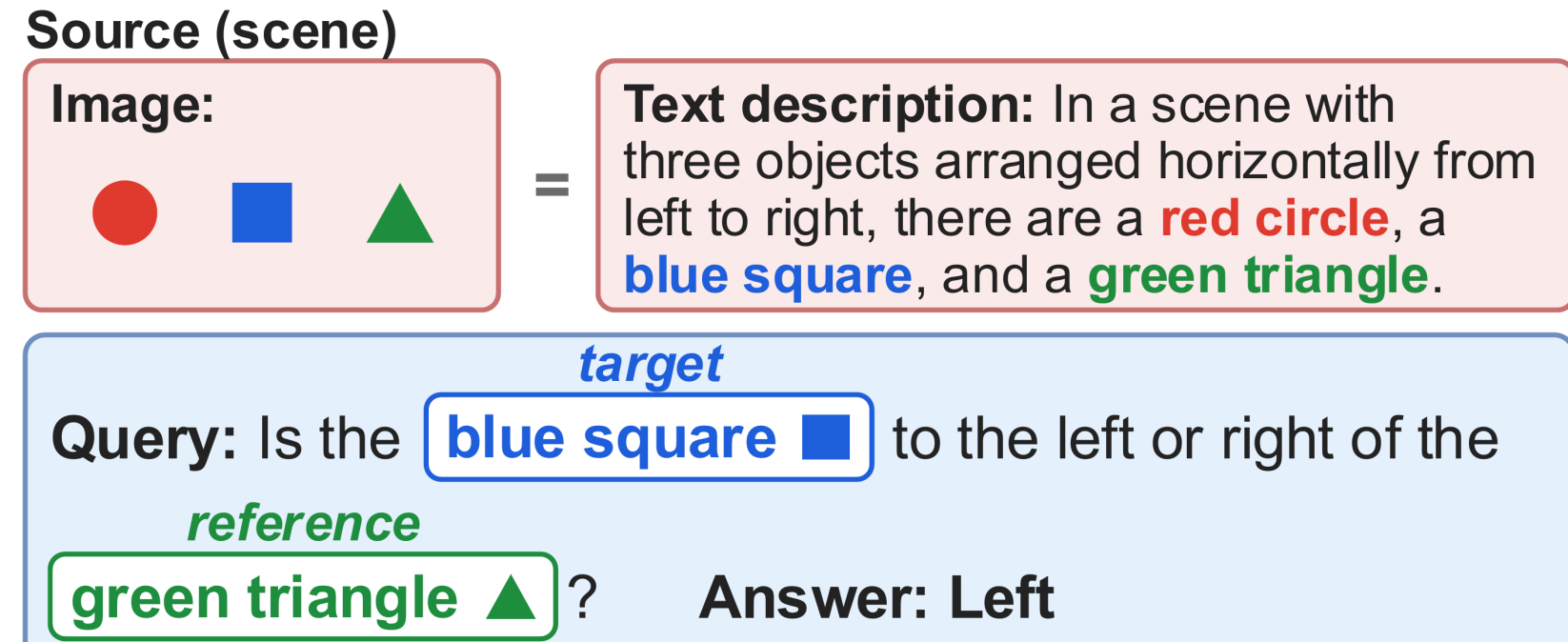


TL;DR We identify complementary contributions of **object-location information** and **query-role representations** to spatial reasoning across visual and textual inputs, and improve accuracy and consistency on held-out natural-image benchmarks through **role-direction steering**.

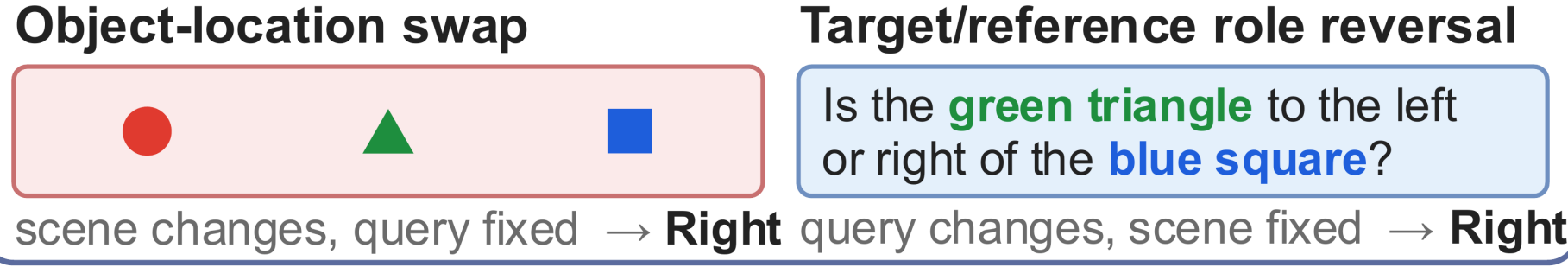
RESEARCH QUESTION How is location information propagated from source to query, and used to compare target/reference roles for predictions?

1 Task, paired consistency tests, and three input conditions

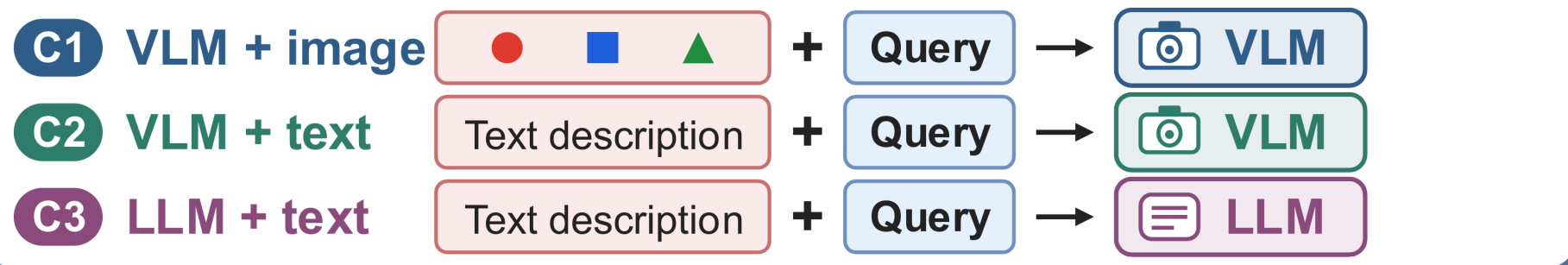
(A) Task example illustration



(B) Paired consistency tests

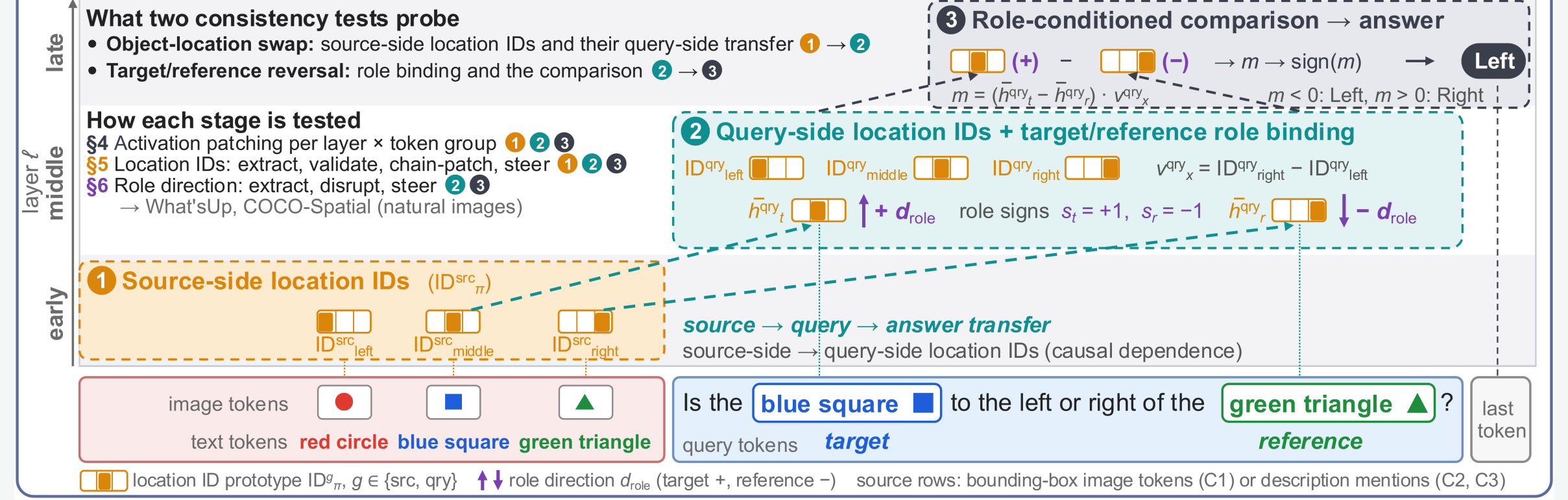


(C) Three matched input conditions

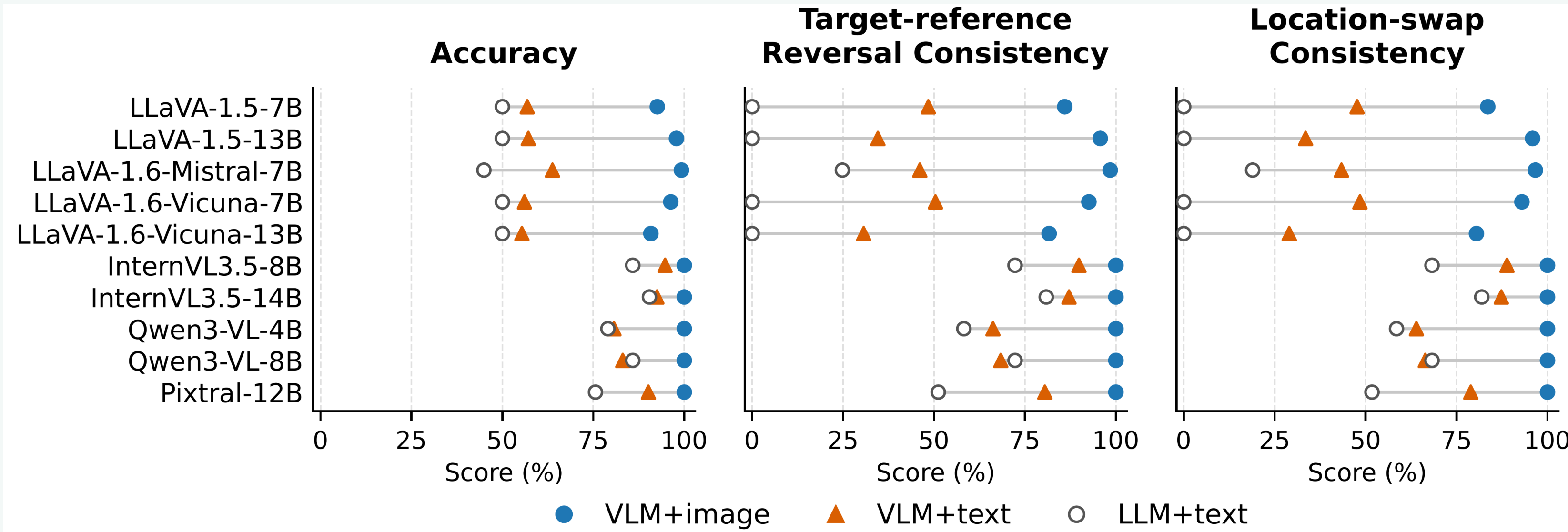


2 Hypothesized mechanism

(D) Hypothesized three-stage mechanism

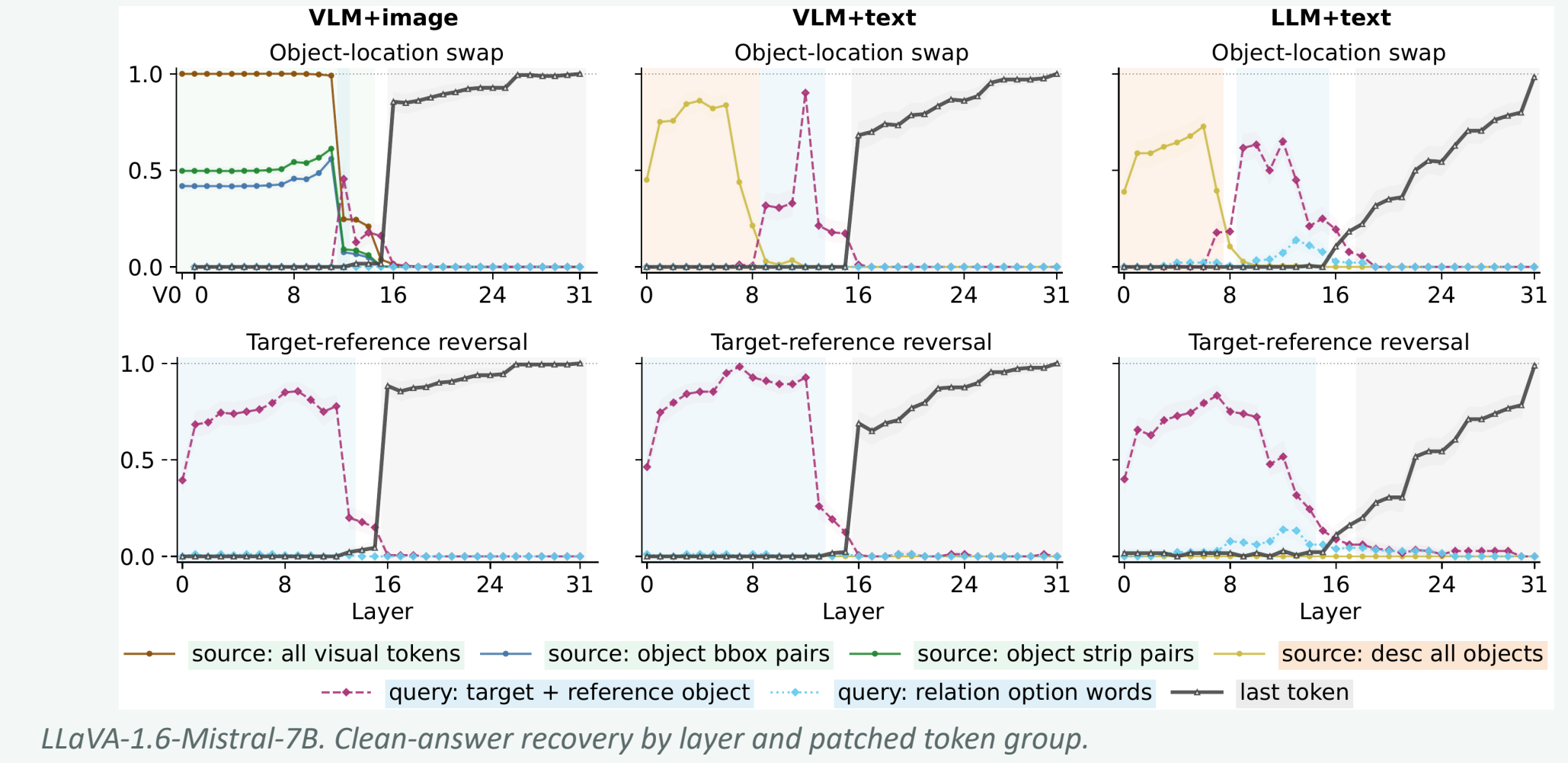


3 High accuracy does not guarantee paired consistency



Synthetic: ten VLMs and their LLM backbones. VLM+image leads on all three metrics; What'sUp shows the same input-condition pattern.

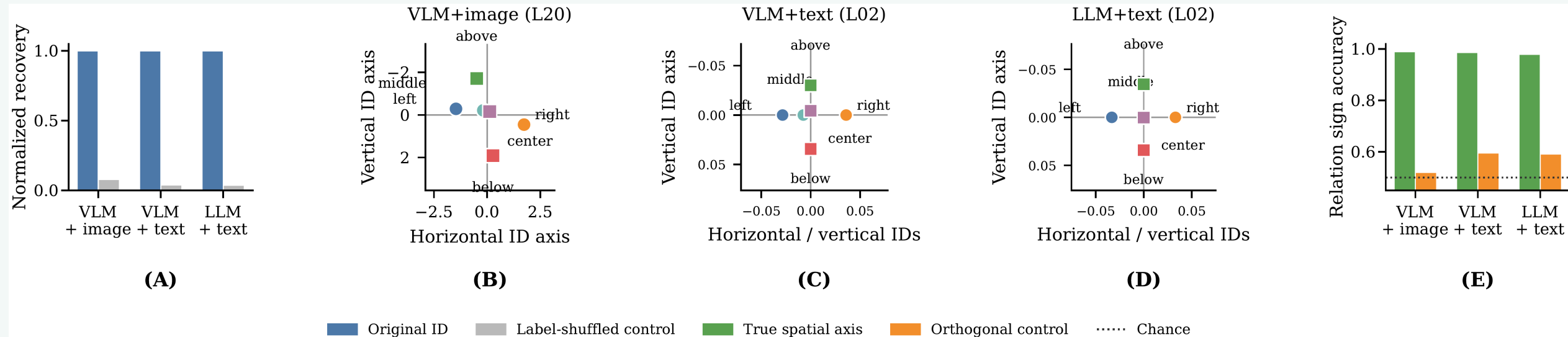
4 Activation patching localizes a staged progression



LLaVA-1.6-Mistral-7B. Clean-answer recovery by layer and patched token group.

Location swap: source (early), query objects (middle), answer state (late).
Role reversal: query-object and final-token patches dominate.

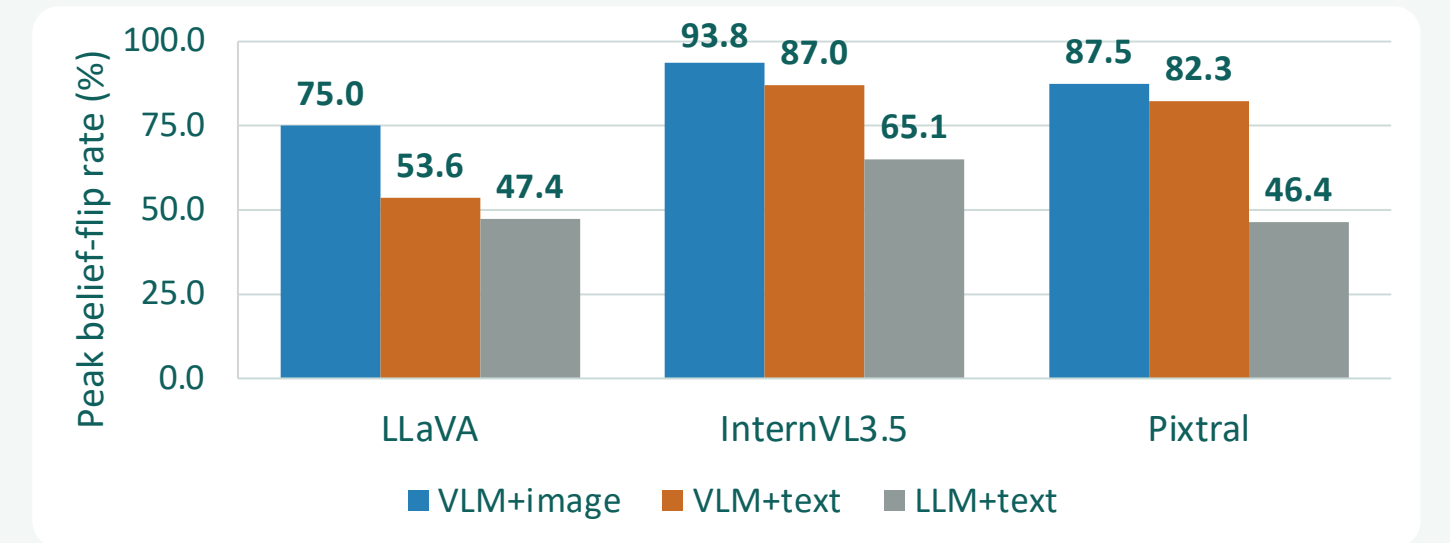
5 Location information is causally linked from source to query to answer



LLaVA-1.6-Mistral-7B. Held-out location decoding, prototype geometry and relation-sign prediction.

Input	Patched source	LLaVA1.6	InternVL3.5	Pixtral
VLM+image	All visual tokens	1.718 / 62.9%	21.660 / 69.1%	3.877 / 51.2%
VLM+image	Object + strip	0.881 / 34.0%	9.691 / 11.6%	1.477 / 5.1%
VLM+text	Object mentions	0.493 / 17.7%	7.912 / 24.3%	0.783 / 13.7%
LLM+text	Object mentions	0.416 / 20.0%	3.586 / 15.6%	0.701 / 10.3%

Swapped-scene source patches: corrupt-directed query-score shift Δm / answer flips (%). All-layer means.



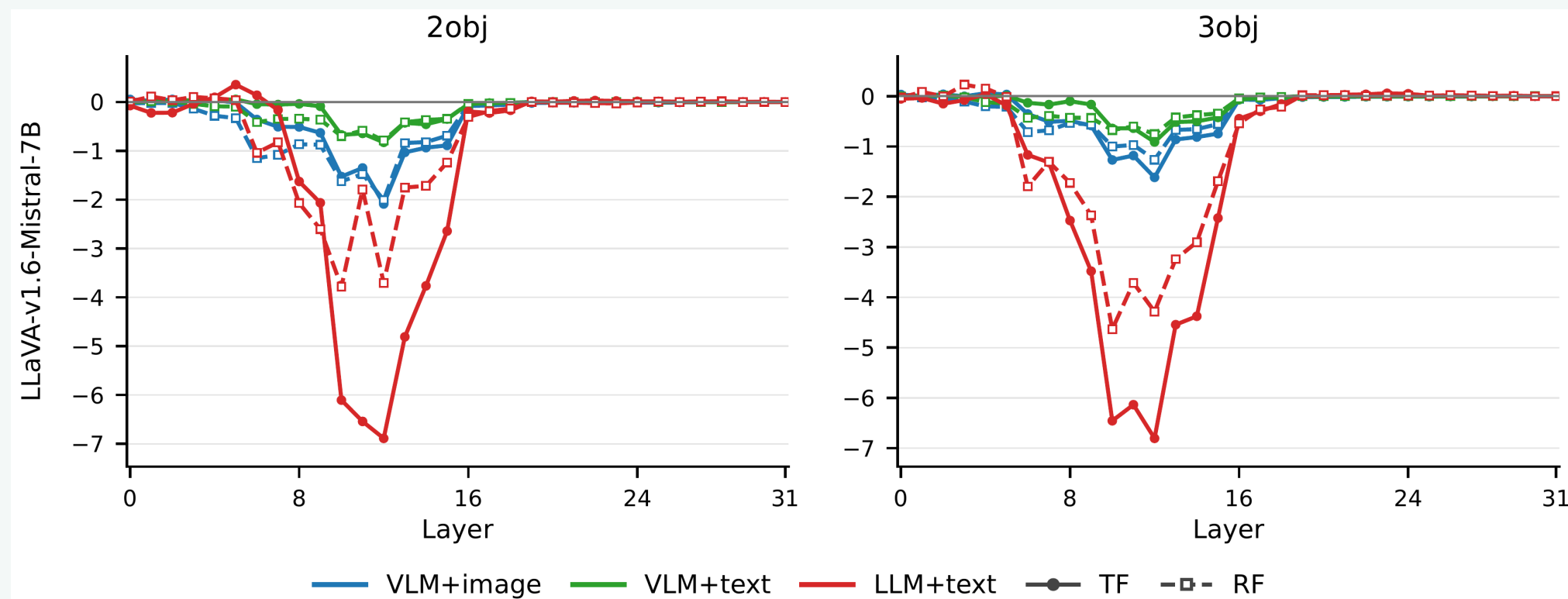
Input unchanged. Peaks across layers ($\alpha = 5$); control directions yield weaker flips.

Location IDs support held-out location prediction and query-side relational comparison, outperforming shuffled-label and orthogonal-axis controls.

Source interventions alter query-side location information and shift answer preferences.

Location-ID steering changes predictions

6 Role directions causally affect predictions and generalize to natural images

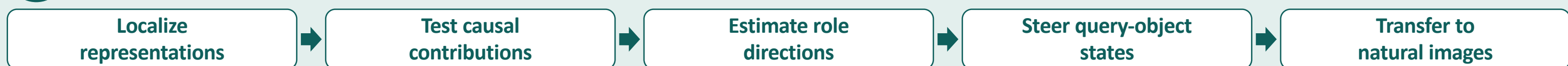


LLaVA-1.6-Mistral-7B. Role disruption lowers the correct-answer margin more than orthogonal controls. TF/RF: target-/reference-first queries.

Model	Input	Accuracy	Role-reversal consistency	Location-swap consistency
LLaVA-1.6	VLM+image	82.2 → 84.0 (+1.8)	70.6 → 73.9 (+3.3)	66.3 → 67.5 (+1.3)
	VLM+text	66.3 → 65.1 (-1.2)	52.2 → 54.9 (+2.7)	39.7 → 39.2 (-0.5)
	LLM+text	63.8 → 67.1 (+3.3)	38.6 → 43.9 (+5.3)	33.5 → 38.2 (+4.7)
InternVL3.5	VLM+image	97.3 → 97.4 (+0.1)	96.1 → 96.6 (+0.5)	94.2 → 94.2 (+0.0)
	VLM+text	87.1 → 89.7 (+2.6)	78.7 → 80.9 (+2.2)	74.1 → 77.8 (+3.7)
	LLM+text	70.8 → 73.2 (+2.4)	46.9 → 50.3 (+3.3)	45.3 → 49.7 (+4.4)
Pixtral	VLM+image	83.5 → 85.0 (+1.6)	69.1 → 72.3 (+3.2)	67.3 → 70.5 (+3.2)
	VLM+text	77.0 → 81.9 (+4.9)	57.8 → 65.7 (+7.8)	54.7 → 64.1 (+9.4)
	LLM+text	74.5 → 78.7 (+4.2)	50.0 → 58.8 (+8.8)	49.6 → 58.0 (+8.3)

What'sUp-A: baseline → steered (change in percentage points). Bold changes: 95% CI excludes zero.

7 Conclusion



Our analyses show that models compare **location information at query-object mentions** according to the objects' **target/reference roles**. **Steering along role directions** estimated on synthetic scenes improves accuracy and paired consistency on natural-image benchmarks **without retraining**.

Actionable Blueprints

- **Counterfactual Consistency Training:** Enforce swap & reversal invariance during SFT / RLHF.
- **Explicit Role-Conditioned Cross-Attention:** Guard against role-assignment corruption in mid-layers.
- **Test-Time Role Direction Steering:** Zero-cost inference-time intervention to boost consistency.